



COMMUNICATION:

The Essence of Science

Facilitating information
exchange among
librarians, scientists,
engineers and students

William D. Garvey



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COMMUNICATION: THE ESSENCE OF SCIENCE



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COMMUNICATION: THE ESSENCE OF SCIENCE

*Facilitating information exchange among librarians, scientists,
engineers and students*

by

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PERGAMON PRESS

OXFORD • NEW YORK • TORONTO • SYDNEY • PARIS • FRANKFURT

UK	Pergamon Press Ltd., Headington Hill Hall, Oxford OX3 0BW, England
USA	Pergamon Press Inc., Maxwell House, Fairview Park, Elmsford, New York 10523, USA
CANADA	Pergamon of Canada, Suite 104, 150 Consumers Road, Willowdale, Ontario M2J 1P9, Canada
AUSTRALIA	Pergamon Press (Aust.) Pty. Ltd., P.O. Box 544, Potts Point, NSW 2011, Australia
FRANCE	Pergamon Press SARL, 24 rue des Ecoles, 75240 Paris, Cedex 05, France
FEDERAL REPUBLIC OF GERMANY	Pergamon Press GmbH, 6242 Kronberg-Taunus, Pferdstrasse 1, Federal Republic of Germany

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First edition 1979

British Library Cataloguing in Publication Data

Garvey, William D.
Communication, the essence of science.
(Pergamon international library)

1. Communication in science
2. Science—Information services
3. Scientific literature

I. Title

029'.9'5 Q223 78-40530

ISBN 0-08-022254-4 Hard cover

ISBN 0-08-023344-9 Flexicover

*Printed and bound at William Clowes & Sons Limited
Beccles and London*

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Preface

Recently, James D. Watson and Francis Crick discussed on the BBC how they made their discovery of the molecular structure of DNA, a discovery which led them, along with Maurice Wilkins, to receive Nobel Prizes in 1962. In the course of their discussion they recalled the time when Maurice Wilkins had in his possession an X-ray photograph of DNA in the B form, which he was trying to keep Linus Pauling, Watson and Crick from seeing until he had a chance to exhaust his own study of the photograph. "But", Crick said, "he *had* to eventually make public the photograph because communication is the essence of science." What I have tried to do in this book is to describe the communication structure of science to the "non-scientist" in such a way that it is apparent why one of the greatest living scientists would say that "communication is the essence of science".

For over a decade, my colleagues and I at The Johns Hopkins University Center for Research in Scientific Communication have been following the behaviour of a large number of scientists associated with several disciplines. Our main interest has been scientific communication which by our definition includes the full spectrum of activities associated with the production, dissemination, and use of information from the time the scientist gets the idea for his research until information about the results of this research is accepted as a constituent of scientific knowledge. One cannot observe this process from the outside as long and intimately as we have without marvelling at the elegance of this complex social system which man consciously created to establish and protect knowledge. It is clear that this marvel has taken place because of the special communication structure which science has developed.

By scientific communication I mean those information-exchange activities which take place mainly among scientists actively involved on the research front. It covers scientific communication from the most informal discussion between two scientists to the formal aspects of scientific communication such as journals, reviews, books, etc. The discussion of these

information-exchange activities is directed toward a group of readers whom I consider to be science's most important future partner. I have referred to this group as "librarians" because my colleagues and I, as basic scientists, have come to recognize the persons who help us in our information needs as being associated with the library. We recognize that they are information specialists who provide us with information services and we are proud to have them associated with us in our endeavour to create and process scientific knowledge. If the new trend to call "librarians" "information specialists" is a more comfortable one for some readers please accept my apologies and substitute your term for my "librarian".

The reason for directing the book's content at this audience is because in recent years it has become increasingly clear that the technology for providing information services to scientists has been less successful than the sophistication of the technology warrants. This problem is not my discovery. I first became aware that many persons involved in information services and technology were acutely aware of this problem when I attended the NATO Advanced Study Institute on "Perspectives in Information Science" held in August 1973 in Aberystwyth, Wales. There was not one session at this 6-day meeting at which this problem was not raised. I would like to offer two quotes from this conference to give some notion of what I have in mind.

Professor Pauline Atherton, from the School of Library Science, Syracuse University, USA, presented a paper entitled "Research in Information Science: An Assessment" in which one of her conclusions was: "The subjects of our research have been system variables more than human variables. I would like to suggest a corrective measure for research in the field. We need to ask and try to answer scientifically, some of the following questions:

- (1) How do people seek information ?
- (2) Why do people seek information ?
- (3) Why do people fail to seek information ?
- (4) Do different types of people seek information in different ways and if so, how can information systems be designed to suit different types of clientele ?
- (5) How do people process information ?

In trying to handle these complex questions, we may have to discard much of the methodology most prevalent in the field today. Inadvertently, perhaps, we have ignored the nearly invisible social processes which affect our work the most, human information processing. Our reliance on system analyses and related techniques has inhibited our study of the more substantive and more difficult aspects of our world of information science, namely the *human being who is processing information*."

My concern at that time could not be put more succinctly. But Professor Atherton was trained as a librarian, presumably trained more in the areas of the "soft" sciences, and perhaps this interpretation of the problem was influenced by sensitivity to the social aspects of the problem. But as I continued to listen to the repeated reference to this problem, it was apparent that concern about it was pervasive among the participants at this meeting. For example, Professor B. C. Brookes, trained as a physical scientist, and now at University College, London, England, presented a quantitatively oriented paper entitled "Statistical Bibliographical and Information Science" in which he said "It seems to me absurd to continue to increase the sophistication of computer technology applied to documentary systems without having a critical look at the material they handle and the use that scientists make of it."

That these concerns, relative to human problems of information production, dissemination, storage, and use, are not just isolated instances flaring up at one conference in the idyllic setting of Wales can be easily demonstrated simply by examining the *Annual Reviews of Information Science and Technology* published during the last decade; the concerns are repeated annually. Bruce A. Alpin's chapter in the 1975 volume raises the general concern: "There continues to be an abundance of publications talking about what could be done, or should be accomplished. What we lack are publications concerned with the question of why particular projects are undertaken: why one particular development philosophy was selected over others considered—if others were, in fact, considered—and what genuine success has occurred . . . we have learned a great deal about what we must implement in terms of computer technology to service daily tasks in the libraries. What we have not learned is how to determine when the job is being done well."

As a social psychologist, I found my private concerns were being articulated by experts in the field of library/information science and technology.

Somehow it seemed to me that we psychologists have not, of all things, communicated with librarians/information specialists. While it is clear to many of us that computer technology is at the point where we must develop a systematic conceptual framework of the social structure in which we expect to employ it, we have not sought the librarians' collaboration in this development. This book was written to try to convince librarians that measures of success or failure of innovations in scientific communication will be measured in terms of social variables and in the hope that collaboration between librarians, scientists, and psychologists in planning, implementing, and evaluating future innovations will become a genuine reality.

The book is introduced by five chapters which give an up-to-date description of the psychologists' point of view of the communication structure of science. These five chapters are followed by an appendix which includes ten articles which have been published by my colleagues and myself. They are intended to give more specific information relative to the general discussion in the text. I am, of course, deeply indebted to these colleagues, for the works reported in these articles have been truly collaborative, and I am especially indebted to Dr Stephen D. Gottfredson, my most recent collaborator. Additionally, I must point out that the works reported in these articles have been supported by grants from the Office of Science Information Service (now the Division of Science Information) of the National Science Foundation (NSF) over the past decade or more. These works have been researched more in the basic nature than the applied, and we have unfailingly received encouragement from the directors of information-science-research programmes at NSF. We are particularly indebted to the current director, Edward Weiss, and former directors, Helen Brownson and Donald Pollock.

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Baltimore, Maryland

CHAPTER 1

The Role of Scientific Communication in the Conduct of Research and the Creation of Scientific Knowledge

For those persons who have not been trained as research scientists or who have not been closely involved in the conduct of scientific research, the description associated with the day-to-day work of scientists sometimes does not match the common stereotype of the dispassionate, objective scientist at work in his secluded laboratory. The process attributed to this stereotype has traditionally been considered to be a matter of applying rational thought to empirical evidence in discovering “truth” about the world. There is no doubt that science arrives at its particular brand of “truth” by logical processing of empirical observations, but any attempt to reduce the process of science solely to its logical principles produces only a partial description of the activities we regard collectively as science.¹

The view of the scientific world as cluttered laboratories in which the scientist, dedicated to intellectual curiosity, pursues his own particular interest in the study of natural phenomena is an oversimplification. He is, not unlike members of other professions, much involved in a competitive social world in which he is motivated by the desire to be successful.² It is the peculiar nature of “success” in science that is perplexing to most non-scientists.

What Is “Success” in Science?

The main effort of individual scientists is manufacturing new information either by describing new data or by formulating new concepts or conceptual integrations of data (theory). In order for these formulations to be successful contributions to science, they must be communicated in

such a form so as to be comprehended and verified by other scientists and then used in providing new ground for further exploration. Thus, communicability becomes a salient feature of a scientific product since its recognition by peers as a unique contribution is essential to establishing a scientist's success in science.

Consequently, throughout a scientist's career he is dependent on the other scientists' evaluation of his work to achieve goals related to the advancement of his career—publication of research, research grants, promotions, etc. Yet, paradoxically, if a scientist is really creative and productive at the cutting edge of the research front, there will probably be only a small, select group (ingroup) of other scientists who can truly understand the significance and appreciate the quality of his work. Each of these few in turn is dependent on the respect of others in the ingroup for recognition of his own success. Thus, we have the situation where a relatively small group of scientists are reinforcing one another for successful contributions while at the same time competing with each other to be the first to proclaim such contributions to the group. Each scientist must enhance his own reputation individually, even though he is committed to the group. His commitment to the group has both an attitudinal and a behavioural component. As an attitude, it is expressed in strong positive feelings about the group, feelings of being a part of it and sharing its goals and successes in the progress of science. As a behaviour, it is expressed in actions which share the burdens as well as benefits of membership, e.g., a willingness to review manuscripts *fairly* in order to maintain the integrity and quality of the group's science.

We have, then, the situation in which the scientist has committed himself to a group with trust in its collective activity, while at the same time he is unwilling to place complete confidence in all individuals in the group in his quest to establish his own reputation. This seeming distrust comes about because being the *first* to make an important scientific contribution is the only way to obtain recognition of his success. In this quest for priority there are stages in the research process when it is better to keep information about his work to himself. If another scientist makes the same contribution at about the same time, he will have to share the success (thus, diluting his own gain in reputation). And if someone makes the contribution only slightly before he does not only will he not get credit for the contribution, he will probably not even be able to publish the results of that work.

Within such a framework of success it becomes clear why the communication structure of science has developed (in the hands of scientists themselves) into the complex and rigorously controlled social system that it is today. And, of course, we begin to understand why scientists are so protective of this system and so resistant to any change proposed. This formalization of the scientific communication process started over three centuries ago when the final step of scientific research customarily became the communication of its results to other scientists; thereafter, science changed its character from a private to a social activity restricted to the world of science.

It would appear then that scientists live in two worlds—a scientific world with its special norms and rigorous communication structure and a separate “outside” world. Scientists have maintained the isolation of their scientific world from the outside world since Galileo established the empirical basis of science and the resistance of this empiricism to outside opinions and dogma. Today the basic-research scientist continues to resist outside influences (e.g., political and social) on the direction of basic research. But in serving the scientist the librarian must live in two worlds at once—in the world of “public” culture in which both he and the scientist work toward the general aims of creating and substantiating knowledge; and in another, more restricted world in which the scientist searches for, creates, and produces *scientific* information. The first world is related to the whole problem of social epistemology and presumably is the culture which the librarian is in charge of organizing and maintaining. The second world is the scientist’s world, with its special communication structure, and the librarian can learn about it either by being trained as a scientist or by studying it as one would study any “foreign” culture.³ Helping the librarian in this study is to a large extent what this book is about.

The Psychology of Scientific Communication

That the same scientists play different roles, have different needs, and respond differently to the same information, is the reason we emphasize the psychological perspective in this book. In taking this perspective we hope to be able to demonstrate the intimate nature between active researchers’

scientific communication behaviour and the behaviour inherent in their scientific creativity.

The psychology of scientific communication emphasizes the interaction between the scientist and his environment (a major element of this environment being other scientists). Each scientist brings to each situation a particular cluster of psychological attributes⁴ (personality, skills, style, experience, habits, etc.), which, combined with specific circumstances in the research process, gives the individual scientist a predisposition to perceive and detect, to assimilate, to associate, etc. what is happening with his research at any given moment. As a consequence, given identical circumstances in the lab, two scientists may perceive them differently, assimilate different information, or apply the same information to fulfil individual needs to advance his particular research. It should be clear why these individual differences exist, but it may not be evident why they are not detrimental to scientific progress. To put it simply, in order to have progress in science we must have variation in observation and selection in interpretation. If we had not had that period during which some scientists saw "oxygen" and interpreted it as phlogiston and others saw an elementary gas and interpreted it as "oxygen", science might have "missed" an important debate which stimulated much research fundamental to modern chemistry.

Hopefully, an understanding of this overall process will provide more general grounds on which the librarian can meet the scientist's needs and advise and assist him in his "continual" pursuit of information.

A major theme throughout this book is the *interactive* nature of scientific communication. Even the scientist's intellectual browsing in his personal searches (e.g., his weekly visits to the library to examine the current periodicals) is interactive in that his style, subjectivity, bias, etc. all play a part in his detection, selection, retention, and use of information encountered in the search. The scientist is not hindered by the fact that his own cluster of these psychological attributes is not identical to that of some other scientist, for the stimulus to scientific creativity rises where there is widespread lack of agreement ranging from the definition of common technical terms to the decision of what constitutes quality in science.⁵ Of course, this subjectivity and idiosyncrasy does not make the librarian's job any simpler; but it is precisely these attributes that allow for unpredictable windings through the maze of scientific knowledge and often result in

unique and valuable findings. Since the scientist is aware of the uncertainty of his information needs and the essential role interaction plays in articulating and satisfying them, it is likely that effective information services will be developed and established through the *joint* efforts of librarians and scientists. To accomplish this the main burden falls on the librarian who must learn the scientist's modes of communication and devise ways of assaying them *in situ*, in order to establish effective means of providing information services. Scientific communication is intimately tied up with the creation of information which evolves slowly. As time passes, however, newly created information is integrated with other information and evolves into more general knowledge, understandable by a wider range of scientists and non-scientists. The sooner librarians can interact with scientists in this process the better they can fulfil their information service functions.

The role of communication among scientists concerning the work they produce, evaluate, disseminate, and utilize in their capacity as researchers and the multiple roles information-exchange plays in the development of scientific knowledge are the subjects of the first article we reprint in Appendix A, "Communication and Information Processing Within Scientific Disciplines: Empirical Findings for Psychology". This article begins with some explanation of why there is an "information crisis" in one science, psychology. The growth of scientific information in psychology is not atypical of science in general, but we emphasized in the article mainly those aspects of special significance to psychology.

The Growth of Scientific Information

Over 300 years ago scientists found that keeping up with one another's current work had become too great a burden to handle by word of mouth and correspondence. As a result, the scientific paper was formalized and distributed by means of scientific journals, the first of these having been established in 1665. Since that time, the number of scientists, the number of scientific papers, and the number of scientific journals has increased steadily and exponentially.

If these things were not exceeding the growth rates of our population or gross national product, the problem of the scientific information deluge